The documentation and process conversion measures necessary to comply with this document shall be completed by 30 April 2023.

INCH-POUND

MIL-PRF-19500/616H
w/AMENDMENT 3
30 January 2023
SUPERSEDING
MIL-PRF-19500/616H
w/AMENDMENT 2
28 January 2022

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, ULTRAFAST
DUAL, COMMON CATHODE OR ANODE CENTER TAP,
TYPES 1N6657 THROUGH 1N6659
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, dual (common anode or common cathode), high voltage, ultrafast power rectifier diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type.

1.2 Package outline. The device package outline is as follows: TO-254AA in accordance with figure 1 for all encapsulated device types.

* 1.3 Maximum ratings (for each leg). Unless otherwise specified T_C = 25°C.

	Types	V_{RWM}	I_{FSM} (1) tp = 8.3 ms	I_F T_C = 100°C (1) (2) (3)	t_{rr} (1)	$R_{\theta JC}$ (1)	$R_{\theta JA}$ (1)	T_{STG} and T_J
		V dc	A (pk)	A dc	ns	°C/W	°C/W	°C
*	1N6657, 1N6657R	90						
*	1N6658, 1N6658R	135	150	15	35	2.3	40	-65 to +200
*	1N6659, 1N6659R	180						

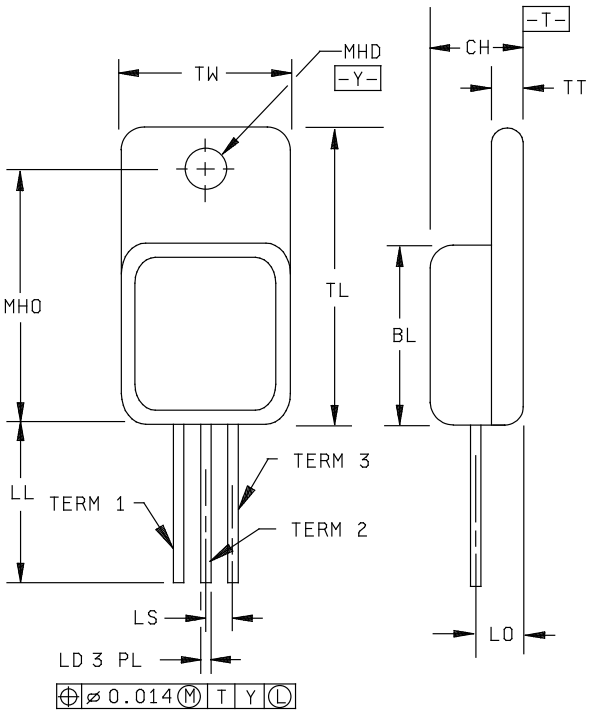
(1) Each individual diode.
(2) Derate linearly at 300 mA/°C from +100°C to +150°C.
(3) Total package current is limited to 30A dc.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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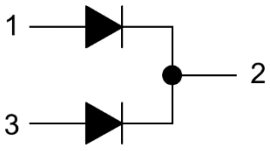


Symbol	Dimensions			
	Inches	Inches	mm	mm
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.530	.550	13.46	13.97
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84

SCHEMATIC

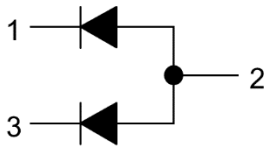
1N6657, 1N6658, 1N6659

TERM 1 = ANODE 1
TERM 2 = CATHODE
TERM 3 = ANODE 2



1N6657R, 1N6658R, 1N6659R

TERM 1 = CATHODE 1
TERM 2 = ANODE
TERM 3 = CATHODE 2



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Protrusion of ceramic eyelets included in dimension LL.
3. All terminals are isolated from case.

FIGURE 1. Dimensions and configuration of three terminal flange mount header TO-254AA package.

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1.4 Primary electrical characteristics. Primary electrical characteristics are not applicable to this specification sheet.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The diodes of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are as follows: "6657", "6658", and "6659".

1.5.3 Suffix letters. The suffix letter "R" indicates that the diode is a reverse polarity version of the stated device.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-254AA) herein. Methods used for electrical isolation of the terminal feed throughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.4.1 Lead formation and finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of MIL-PRF-19500.

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

* Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
1a	Required	Not required
1b	Required	Required (JANTXV only)
3a	Required	Required
3b	Surge current (see 4.3.1)	Surge current (see 4.3.1)
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
6	Not applicable	Not applicable
9	Not applicable	Not applicable
10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	(see 4.3.3), t = 240 hours	(see 4.3.3), t = 48 hours
* 13	Subgroup 2 and 3 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±0.1 V (pk); ΔI_{R1} = ±2 µA dc or 100 percent from the initial value; whichever is greater.	Subgroup 2 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±0.1 V (pk); ΔI_{R1} = ±2 µA dc or 100 percent from the initial value; whichever is greater.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Surge current. The surge current test shall be performed in accordance with condition A, method 4066 of MIL-STD-750. I_O = 0; $V_{RM(w)}$ = 0; I_{FSM} = see 1.3; six surges per leg; T_A = 25°C, t_P = 8.3 ms, one minute minimum time between surges.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081, as applicable, of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). See table II, group E, subgroup 4 herein.

* 4.3.3 Power Burn-in conditions. Burn-in conditions shall be in accordance with test condition A of method 1038 of MIL-STD-750 and as follows: T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. V_R = 0.9 of rated V_{RWM}. Adjust I_O or I_F to achieve T_J =150°C minimum.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions of tables E-VIA (JANS) and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. See table III herein for delta limits when applicable.

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4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. I_F or I_O = 1.25 to 10 A.
B5	1027	For irradiated devices, include t_{rr} as an end-point measurement.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. I_F or I_O = 1.25 to 10 A. For irradiated devices, include t_{rr} as an end-point measurement.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. See table III herein for delta limits when applicable.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 pounds (4.54 kg), t = 15 seconds.
C5	4081	$R_{\theta JC}$ = 2.3°C/W.
C6	1037	T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. I_F or I_O = 1.25 to 10 A, for irradiated devices, include t_{rr} as an end-point measurement.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and table II herein. See table III herein for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectifier current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.

* 4.5.3 Scope display evaluation. Th e reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 µA/division and 20 to 50 V/division. Reverse current over the knee shall be at least 500 µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
* Thermal impedance 2/	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
* Breakdown voltage	4022	I_{BR} = 50 µA dc; pulsed (see 4.5.1)	V_{BR}			V dc
* 1N6657, 1N6657R				100		
* 1N6658, 1N6658R				150		
* 1N6659, 1N6659R				200		
Forward voltage	4011	Condition B, I_F = 10 A (pk); pulsed (see 4.5.1)	V_{F1}		1.0	V dc
Forward voltage	4011	Condition B, I_F = 20 A (pk); pulsed (see 4.5.1)	V_{F2}		1.2	V dc
* Reverse current leakage	4016	Condition A or B; V_R = rated, V_{RWM} (see 1.3); pulsed (see 4.5.1)	I_{R1}		10	µA dc
Subgroup 3						
High temperature operation		T_C = +100°C				
* Reverse current leakage	4016	Condition A or B; V_R = rated V_{RWM} (see 1.3); pulsed (see 4.5.1)	I_{R2}		1.0	mA dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	Condition B, I_F = 10 A (pk); pulsed (see 4.5.1)	V_{F3}		1.15	V dc
Subgroup 4						
Scope display evaluation		See 4.5.3				
Reverse recovery time	4031	Condition B; I_F = 1A dc, I_R = 1 A, I_{RR} = 100 mA	t_{rr}		35	ns
Junction capacitance	4001	V_R = 10 V dc, f = 1 MHz, V_{SIG} = 50 mV (p-p) (max)	C_J		150	pF

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, mounting conditions in accordance with test method 1026 of MIL-STD-750, T_A = +25°C, I_{FSM} = (see 1.3), I_O = 0; $V_{RM(w)}$ = 0; six surges; T_A = 25°C, t_P = 8.3 ms, one minute minimum time between surges.				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4 and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles, condition C, - 55°C to +175°C.	
Hermetic seal	1071		
Electrical measurements		See table I, subgroup 2 and table III, steps 1 and 2. For irradiated devices, include t_{rr} as an end-point measurement.	
Subgroup 2			45 devices, c = 0
Blocking life	1048	1,000 hours, V_R = 0.9 of rated V_{RWM}.	
Electrical measurements		See table I, subgroup 2 herein, except for thermal impedance and table III, steps 1 and 2. For irradiated devices, include t_{rr} as an end-point measurement.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices, c = 0
Barometric pressure, reduced (altitude operation)	1001	V_R = rated V_R (see 1.3); pressure = 33 mm Hg; t = 1 minute (minimum), R_{ISO} = 2.0 x 10^6 ohm minimum, I_R = I_{R1}	
Subgroup 8			22 devices, c = 0
Forward surge	4066	Condition A, mounting conditions in accordance with test method 1026 of MIL-STD-750, T_A = +25°C, I_{FSM} = (see 1.3), I_O = 0; $V_{RM(w)}$ = 0; six surges; T_A = 25°C, tp = 8.3 ms, one minute minimum time between surges.	
Electrical measurements		See table I, subgroup 2 herein.	

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* TABLE III. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits
		Method	Conditions		
1.	Forward voltage change	4011	Condition B, I_{FM} = 10A, pulsed (see 4.5.1)	ΔV_{F1}	±0.1 V (pk) from previous measured value
* 2.	Reverse current change	4016	Condition A or B, V_R = 90 percent of rated V_{RWM} (see 1.3)	ΔI_{R1}	±2 µA dc or 100 percent from initial value, whichever is greater.

*

1/ The delta measurements taken during group B inspections for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 4, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 5, the measurements of steps 1 and 2 of this table shall also be taken.

2/ The delta measurements taken during group B inspections for quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken.

3/ The delta measurements taken during group C inspections for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows: In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken.

4/ The delta measurements taken during group E inspections for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows: In addition to the measurements specified for subgroups 1 and 2, the measurements of steps 1 and 2 of this table shall also be taken.

5/ Devices which fail the limits of table I herein shall not be acceptable.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

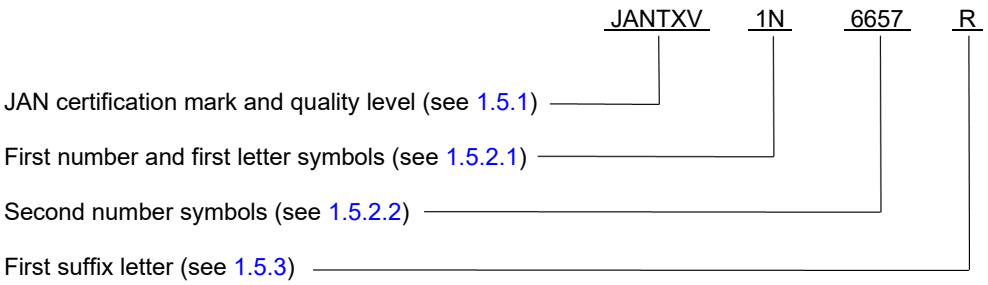
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN1N6657	JANTX1N6657	JANTXV1N6657	JANS1N6657
JAN1N6657R	JANTX1N6657R	JANTXV1N6657R	JANS1N6657R
JAN1N6658	JANTX1N6658	JANTXV1N6658	JANS1N6658
JAN1N6658R	JANTX1N6658R	JANTXV1N6658R	JANS1N6658R
JAN1N6659	JANTX1N6659	JANTXV1N6659	JANS1N6659
JAN1N6659R	JANTX1N6659R	JANTXV1N6659R	JANS1N6659R

6.6 Interchangeability information. Devices covered by this document are configured as a dual TO-254 package version of MIL-PRF-19500/478, which is a stud package version.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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